

Program 10 : Deploy an application using k8s deployment and service

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ ls
deployment.yaml  service.yaml
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ minikube start --cpus=2 memory=4096
🐳 minikube v1.37.0 on Ubuntu 22.04
👉 Using the docker driver based on existing profile
🌟 Starting "minikube" primary control-plane node in "minikube" cluster
🚢 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌞 Enabled addons: storage-provisioner, default-storageclass
🏡 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ █
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment configured
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ kubectl sply -f service.yaml
error: unknown command "sply" for "kubectl"
```

Did you mean this?

```
    apply
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ kubectl apply -f service.yaml
service/nginx-service created
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ █
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-h8wjc   1/1     Running   3 (2m31s ago)    4h54m
nginx-deployment-6f9664446b-xpszz   1/1     Running   3 (2m31s ago)    4h54m
program-8                            1/1     Running   2 (2m21s ago)    31m
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ kubectl get nodes
NAME      STATUS    ROLES          AGE      VERSION
minikube   Ready     control-plane  4h55m    v1.34.0
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ kubectl get svc
NAME            TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes      ClusterIP     10.96.0.1     <none>         443/TCP           4h58m
nginx-service    NodePort      10.103.17.89 <none>         80:31571/TCP      2m59s
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:31571

```
🚀 Opening service default/nginx-service in default browser...
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg10$ Opening in existing browser session.
```

